

PSET 9: Classical statistical inference

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Properties of estimators

- Let $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$ and let $\hat{\lambda} = \frac{\sum_{i=1}^n X_i}{2n}$. Find the bias, standard error, and MSE of this estimator.¹
- Let $X_1, \dots, X_n \sim \text{Uniform}(0, \theta)$ and let $\hat{\theta} = 3\bar{X}_n$. Find the bias, standard error, and MSE of this estimator. (Recall that if $X \sim \text{Uniform}(0, \theta)$, then $E[X] = \theta/2$ and $\text{Var}(X) = \theta^2/12$.)²

Problem 2: Social newsing

A poll conducted in 2025 found that 12% of U.S. adults get at least some news on TikTok. The standard error for this estimate was 2.4%, and a normal distribution may be used to model the sample proportion.³

Construct a 95% confidence interval for the fraction of U.S. adults who get some news on TikTok, and interpret the confidence interval in context.

Problem 3: (T/F) Social newsing

Identify the following statements as true or false using the information from Problem 2. Provide an explanation to justify each of your answers. Explain in approximately 75 words.

- The data provide statistically significant evidence that more than 16% of U.S. adults get some news through TikTok. Use a significance level of $\alpha = 0.05$.
- Since the standard error is 2.4%, we can conclude that 97.6% of all U.S. adults were included in the study.
- If we want to reduce the standard error of the estimate, we should collect less data.

¹Inspired by Wasserman 6.6.1

²Inspired by Wasserman 6.6.3

³Inspired by OI 4.8 and 4.10

Problem 4: Dating on college campuses

A survey conducted on a reasonably random sample of 200 undergraduates asked, among many other questions, about the number of exclusive relationships these students have been in. The sample average is 3.2 with a standard deviation of 2.17.⁴

Estimate the average number of exclusive relationships undergraduate students have been in using the Normal distribution and a 99% confidence interval, and interpret this interval in context. Would a 95% interval be wider or narrower than the one you found? Explain in one sentence.

Problem 5: Statistical significance

Determine whether the following statement is true or false, and explain your reasoning: “With large sample sizes, even small differences between the null value and the point estimate can be statistically significant.”⁵

Problem 6: Sleep deprivation

New York is known as “the city that never sleeps”. Eight hours is the standard nightly recommendation. A random sample of 25 New Yorkers were asked how much sleep they get per night; statistical summaries of these data are shown below. Do these data provide strong evidence that the average amount New Yorkers sleep differs from 8 hours a night? Use a *two-sided* test with $\alpha = 0.05$, per our course convention.⁶

n	$\bar{x}$	s	min	max
25	7.63	0.87	6.17	9.78

Critical values of the t distribution with 24 degrees of freedom:

two-tailed α	0.10	0.05	0.02	0.01
t^*	1.711	2.064	2.492	2.797

- Write the hypotheses in symbols and in words.
- Calculate the test statistic, T , and state the degrees of freedom.
- Using the table above, bound the p-value between two of the listed α levels and interpret it in this context.
- What is the conclusion of the hypothesis test?
- Suppose that New Yorkers really do sleep 8 hours a night on average. What type of error would your conclusion in part (d) represent? Describe what the other type of error would look like in this setting.

⁴Inspired by OI 4.15

⁵Inspired by OI 4.47

⁶Inspired by OI 5.7

Problem 7: Interpreting public opinion polls

On June 28, 2012 the U.S. Supreme Court upheld the much debated 2010 healthcare law, declaring it constitutional. A Gallup poll released the day after this decision indicates that 46% of 1,012 Americans agree with this decision. At a 95% confidence level, this sample has a 3% margin of error. Based on this information, determine if the following statements are true or false, and explain your reasoning.⁷

- a. We are 95% confident that between 43% and 49% of Americans in this sample support the decision of the U.S. Supreme Court on the 2010 healthcare law.
- b. We are 95% confident that between 43% and 49% of Americans support the decision of the U.S. Supreme Court on the 2010 healthcare law.
- c. If we considered many random samples of 1,012 Americans, and we calculated the sample proportions of those who support the decision of the U.S. Supreme Court, 95% of those sample proportions will be between 43% and 49%.
- d. The margin of error at a 90% confidence level would be higher than 3%.

Problem 8: Power and sample size

Suppose you have two samples: one with a sample mean of 12, a standard deviation of 2.13, and n_1 of 87; and a second with a sample mean of 12.8, a standard deviation of 3.25, and n_2 of 89. Throughout, use a two-sided *Wald test* of $H_0 : \delta = 0$ versus $H_1 : \delta \neq 0$, where $\delta = \mu_1 - \mu_2$.

- a. Is it statistically significant?
- b. Now suppose the true difference in means is in fact 0.8. Using the power function for the Wald test from class, find the power of this test at $\alpha = 0.05$. What does your answer say about how much to trust your conclusion in part (a)?
- c. What is the smallest n_2 you could have if you want to reject the null hypothesis at $\alpha < 0.05$?
- d. Darn it! Your n_2 is actually only 85! What should your goal be: smaller sample deviation by 10% or larger sample by 10? Explain your reasoning and provide mathematical support.

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.

⁷Inspired by OI 6.6

Survey questions

1. How prepared do you feel for the final exam? not prepared 1 2 3 4 5 very prepared
2. Which content area of the course was the easiest for you? Why? Please be specific.
3. Which content area of the course was the most challenging for you? Why? Please be specific.